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Course name: Data Structures and Algorithm

course code: BCSE202P

Lab Assessment ONE

Part A

For uploading in VTOP in single PDF:

- 1) Code
- 2) Code screenshot
- 3) Output screenshot
- 4) Logic explanation for each function.

For the both problem 1 and problem 2, Consider n=5 to store five elements in the stack, explain function with each iteration.

Questions

Program on stack:

1) Write a C program to perform the following stack operation for the integer data type:

- A. Push()
- B. Pop()
- C. Print the top of the element
- D. Print all the element of the stack
- E. Quit.

Print all the above functions in the console window. Ask the user to select one of the functions to perform the operation. For example, if the user enter choice 1 – then he can push the element to the stack.

If the user again enter the choice 1 – then he can push one more element to the stack.

If the user enter the choice 3 – program has to print the top of the element in the stack.

Code Part

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  #define MAX 5
5
6  int arr[MAX];
7  int top = -1;
8
9  // Push
10 void push(int value) {
11     if(top == MAX - 1) {
12         printf("Stack Overflow\n");
13     } else {
14         top++;
15         arr[top] = value;
16     }
17 }
18
19 // Pop
20 void pop() {
21     if(top == -1) {
22         printf("Stack Underflow\n");
23     } else {
24         printf("Popped element is: %d\n", arr[top]);
25         top--;
26     }
27 }
28
29 // Peek / Top element
30 void peek() {
31     if(top == -1) {
32         printf("Stack is empty\n");
33     } else {
34         printf("Top element is: %d\n", arr[top]);
35     }
36 }
37
38 // Print all elements
39 void print_all() {
40     if(top == -1) {
41         printf("Stack is empty\n");
42     } else {
43         printf("Stack elements are:");
44         for(int i = 0; i <= top; i++) {
45             printf(" %d", arr[i]);
46         }
47         printf("\n");
48     }
49 }
50
51 // IsEmpty check
52 void isEmpty() {
53     if(top == -1) {
54         printf("The stack is empty\n");
55     } else {
56         printf("The stack is not empty\n");
57     }
58 }
59
```

```

60 int main() {
61     int choice, value;
62
63     printf("Welcome to Stack Operations\n");
64     printf("1. Push an element\n");
65     printf("2. Pop an element\n");
66     printf("3. Print the top element\n");
67     printf("4. Print all the elements\n");
68     printf("5. Exit\n");
69
70     while(1) {
71         printf("\nEnter your choice: ");
72         scanf("%d", &choice);
73
74         switch(choice) {
75             case 1:
76                 printf("Enter the value to push: ");
77                 scanf("%d", &value);
78                 push(value);
79                 break;
80             case 2:
81                 pop();
82                 break;
83             case 3:
84                 peek();
85                 break;
86             case 4:
87                 print_all();
88                 break;
89             case 5:
90                 printf("Exiting program.\n");
91                 exit(0);
92             default:
93                 printf("Invalid choice. Try again.\n");
94         }
95     }
96
97     return 0;
98 }

```

Output

```
Welcome to Stack Operations
1. Push an element
2. Pop an element
3. Print the top element
4. Print all the elements
5. Exit

Enter your choice: 1
Enter the value to push: 1

Enter your choice: 1
Enter the value to push: 2

Enter your choice: 1
Enter the value to push: 3

Enter your choice: 1
Enter the value to push: 4

Enter your choice: 1
Enter the value to push: 5

Enter your choice: 4
Stack elements are: 1 2 3 4 5

Enter your choice: 2
Popped element is: 5

Enter your choice: 3
Top element is: 4

Enter your choice: 5
Exiting program.
[1] + Done                                     "/usr/bin/gdb" --interpreter=mi --tty=${DbgTerm
ronit-mexson@ronit-mexson-Inspiron-15-5510:~/Coding/Coding/Youtube/C++(DSA)$
```

2. Write a C program to perform the following stack operation for the character data type:

- A. Push()
- B. Pop()
- C. Print the top of the element
- D. Print all the element of the stack
- E. Quit.

Print all the above functions in the console window. Ask the user to select one of the functions to perform the operation. For example, if the user enter choice 1 – then he can push the element to the stack.

If the user again enter the choice 1 – then he can push one more element to the stack.

If the user enter the choice 3 – program has to print the top of the element in the stack.

Code

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  #define MAX 5
5
6  char arr[MAX];
7  int top = -1;
8
9  // Push
10 void push(char value) {
11     if(top == MAX - 1) {
12         printf("Stack Overflow\n");
13     } else {
14         top++;
15         arr[top] = value;
16     }
17 }
18
19 // Pop
20 void pop() {
21     if(top == -1) {
22         printf("Stack Underflow\n");
23     } else {
24         printf("Popped element is: %c\n", arr[top]);
25         top--;
26     }
27 }
28
29 // Peek / Top element
30 void peek() {
31     if(top == -1) {
32         printf("Stack is empty\n");
33     } else {
34         printf("Top element is: %c\n", arr[top]);
35     }
36 }
37
```

```

38 // Print all elements
39 void print_all() {
40     if(top == -1) {
41         printf("Stack is empty\n");
42     } else {
43         printf("Stack elements are:");
44         for(int i = 0; i <= top; i++) {
45             printf(" %c", arr[i]);
46         }
47         printf("\n");
48     }
49 }
50
51 // IsEmpty check
52 void isEmpty() {
53     if(top == -1) {
54         printf("The stack is empty\n");
55     } else {
56         printf("The stack is not empty\n");
57     }
58 }
59

```

```

60 int main() {
61     int choice;
62     char value;
63
64     printf("Welcome to Stack Operations\n");
65     printf("1. Push an element\n");
66     printf("2. Pop an element\n");
67     printf("3. Print the top element\n");
68     printf("4. Print all the elements\n");
69     printf("5. Exit\n");
70
71     while(1) {
72         printf("\nEnter your choice: ");
73         scanf("%d", &choice);
74
75         switch(choice) {
76             case 1:
77                 printf("Enter the value to push: ");
78                 scanf(" %c", &value);
79                 push(value);
80                 break;
81             case 2:
82                 pop();
83                 break;
84             case 3:
85                 peek();
86                 break;
87             case 4:
88                 print_all();
89                 break;
90             case 5:
91                 printf("Exiting program.\n");
92                 exit(0);
93             default:
94                 printf("Invalid choice. Try again.\n");
95         }
96     }
97
98     return 0;
99 }

```

Output

Welcome to Stack Operations

1. Push an element
2. Pop an element
3. Print the top element
4. Print all the elements
5. Exit

Enter your choice: 1

Enter the value to push: 1

Enter your choice: 1

Enter the value to push: 2

Enter your choice: 1

Enter the value to push: 3

Enter your choice: 1

Enter the value to push: 4

Enter your choice: 1

Enter the value to push: 5

Enter your choice: 4

Stack elements are: 1 2 3 4 5

Enter your choice: 2

Popped element is: 5

Enter your choice: 3

Top element is: 4

Enter your choice: 5

Exiting program.

[1] + Done

"/usr/bin/gdb" --interpreter=mi --tty=\${DbgTerm}

ronit-mexson@ronit-mexson-Inspiron-15-5510:~/Coding/Coding/Youtube/C++(DSA)\$